

Bachelor Project: Build an FTP Client in Java

Goal:

Implement a FTP Client using Java that can interact with a public FTP server (e.g., ftp.gnu.org).

1. Allowed Libraries (Strict Restriction)

Only the following Java packages are allowed:

- java.io.*
- java.net.*
- java.util.*

No third-party libraries or additional Java packages are permitted.

2. Required Features

- Connect to FTP server (default port 21)
- Anonymous login and custom username/password login
- pwd (PWD command)
- cd (CWD command)
- ls using PASV + LIST
- get using PASV + RETR (binary mode)
- put using PASV + STOR (binary mode)
- delete (DELE command)
- mkdir (MKD command)
- rmdir (RMD command)
- quit (QUIT command)

3. Technical Requirements

- Must correctly handle control connection and separate data connection.
- Must correctly parse FTP reply codes, including multiline responses.
- Must properly close sockets and streams.
- Must handle errors gracefully.

4. Submission

1. Source code (.java files)
2. README with build/run instructions
3. Short report (3–8 pages) explaining architecture and protocol flow, showing test commands

You must submit a zip file including 3 above files into Moodle website.

5. Grading Rubric (100 Points)

- Open/Connect – 5 points - Login (USER/PASS) – 10 points - PWD + CWD – 10 points - PASV + LIST – 10 points - Download (RETR) – 10 points - Upload (STOR) – 10 points - Simple Graphic User Interface (GUI)- 10 points.	- Remote file operations – 5 points - Response parsing (including multiline) – 8 points - Resource management – 6 points - Error handling & robustness – 6 points - Documentation quality – 6 points - Code structure & clarity – 4 points
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This is an example of simple GUI for FTP Client. If you implement this function, you can use two extra libraries: Swing and JavaFX

